

Q1. UV–Visible Spectrophotometer

Definition, Principle (Beer–Lambert law), Construction, Working and Applications explained.

Q2. Demineralization of Water

Ion-exchange principle, process, reactions, advantages and applications.

Q3. Liquid Crystal Display (LCD)

Definition, principle, construction, working, advantages and applications.

Q4. Synthesis of CuO by Precipitation Method

Aim, principle, procedure, reactions and result.

Q5. Use of Gold Nanomaterials in Cancer Treatment

Introduction, mechanisms, applications, advantages and conclusion.

Q6. OLEDs

Definition, principle, construction, working, advantages and applications.

Q7. Titration Curves

HCl vs strong base and HCl vs weak base with explanations.

Q8. Numerical on COD

Definition, principle, formula and calculation format.

Q9. Selective Catalytic Reduction (SCR)

Definition, principle, process, reactions, advantages and applications.

Q10. Sewage Treatment – Activated Sludge Method

Definition, principle, process, advantages, disadvantages and applications.